

SQL Notes

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1 About the notes

These notes cover the basics of SQL using the Northwind database in SQLite. This document is intended to provide a quick reference for SQL queries and operations, and is used in my 1st year information technologies course at Istanbul

University. The notes include examples of SELECT, UPDATE, DELETE, and INSERT INTO statements, as well as the use of the LIMIT clause and various filtering techniques. The notes also include limited examples of JOIN operations and summary statistics using some aggregate functions (e.g., COUNT, SUM, AVG).

2 The Database Schema

The database is `Northwind.sqlite`, which is available to download at <https://github.com/jbytecode/sampledbs/tree/main/sqlite>. This database contains the following tables:

- Categories
- Customers
- Employees
- Order Details
- Orders
- Products
- Shippers
- Suppliers
- Regions
- Territories
- EmployeeTerritories

3 LIMIT

The LIMIT clause is used to specify the maximum number of records to return from a query. It is often used in conjunction with the SELECT statement to limit the number of rows returned.

4 Comments

In SQL, we can define single-line comments using two hyphens (--).

```
-- Get first 5 Customers from Sweden
SELECT
ContactName, ContactTitle, Country, City
FROM Customers
```

```

WHERE
Country = 'Sweden'
LIMIT 5;

```

ContactName	ContactTitle	Country	City
Christina Berglund	Order Administrator	Sweden	Luleå
Maria Larsson	Owner	Sweden	Bräcke

5 SELECT

5.1 Basic Select

```

SELECT CompanyName, ContactName, City FROM Customers LIMIT 5;

```

CompanyName	ContactName	City
Alfreds Futterkiste	Maria Anders	Berlin
Ana Trujillo Emparedados y helados	Ana Trujillo	México D.F.
Antonio Moreno Taquería	Antonio Moreno	México D.F.
Around the Horn	Thomas Hardy	London
Berglunds snabbköp	Christina Berglund	Luleå

5.2 Renaming Columns (Fields) by AS

```

SELECT
    ProductName,
    UnitPrice
FROM Products
LIMIT 5;

```

ProductName	UnitPrice
Chai	18
Chang	19
Aniseed Syrup	10
Chef Anton's Cajun Seasoning	22
Chef Anton's Gumbo Mix	21.35

```

SELECT
    ProductName as 'Name',
    UnitPrice as 'Price'
FROM Products
LIMIT 5;

```

Name	Price
Chai	18
Chang	19
Aniseed Syrup	10
Chef Anton's Cajun Seasoning	22
Chef Anton's Gumbo Mix	21.35

5.3 Filter by City

```
SELECT CompanyName, ContactName, City FROM Customers
WHERE City = 'Berlin'
LIMIT 5;
```

CompanyName	ContactName	City
Alfreds Futterkiste	Maria Anders	Berlin

5.4 Filter by Country and Order by Company Name

```
SELECT CompanyName, ContactName, City, Country FROM Customers
WHERE Country = 'UK'
ORDER BY CompanyName DESC
LIMIT 5;
```

CompanyName	ContactName	City	Country
Seven Seas Imports	Hari Kumar	London	UK
North/South	Simon Crowther	London	UK
Island Trading	Helen Bennett	Cowes	UK
Eastern Connection	Ann Devon	London	UK
Consolidated Holdings	Elizabeth Brown	London	UK

5.5 Filter by Country and Order by City and Country

```
SELECT CompanyName, ContactName, City, Country FROM Customers
WHERE Country IN ('UK', 'Germany')
ORDER BY Country ASC, City ASC
LIMIT 15;
```

CompanyName	ContactName	City	Country
Drachenblut Delikatessen	Sven Ottlieb	Aachen	Germany
Alfreds Futterkiste	Maria Anders	Berlin	Germany
Königlich Essen	Philip Cramer	Brandenburg	Germany
QUICK-Stop	Horst Kloss	Cunewalde	Germany
Lehmanns Marktstand	Renate Messner	Frankfurt a.M.	Germany
Ottilies Käseladen	Henriette Pfalzheim	Köln	Germany
Morgenstern Gesundkost	Alexander Feuer	Leipzig	Germany
Blauer See Delikatessen	Hanna Moos	Mannheim	Germany
Frankenversand	Peter Franken	München	Germany
Toms Spezialitäten	Karin Josephs	Münster	Germany
Die Wandernde Kuh	Rita Müller	Stuttgart	Germany
Island Trading	Helen Bennett	Cowes	UK
Around the Horn	Thomas Hardy	London	UK
B's Beverages	Victoria Ashworth	London	UK
Consolidated Holdings	Elizabeth Brown	London	UK

5.6 Filter by Phone Number

```
SELECT CompanyName, Phone
FROM Customers
WHERE Phone LIKE '(91)%'
LIMIT 5;
```

CompanyName	Phone
Bólido Comidas preparadas	(91) 555 22 82
FISSA Fabrica Inter. Salchichas S.A.	(91) 555 94 44
Romero y tomillo	(91) 745 6200

5.7 Filter by Phone Number Ending with 82

```
SELECT CompanyName, Phone
FROM Customers
WHERE Phone LIKE '%82'
LIMIT 5;
```

CompanyName	Phone
Bólido Comidas preparadas	(91) 555 22 82
Consolidated Holdings	(171) 555-2282
Godos Cocina Típica	(95) 555 82 82
Gourmet Lanchonetes	(11) 555-9482

5.8 Filter by Phone Number Containing Hyphen

```
SELECT CompanyName, Phone
FROM Customers
WHERE Phone LIKE '%-%'
LIMIT 5;
```

CompanyName	Phone
Alfreds Futterkiste	030-0074321
Ana Trujillo Emparedados y helados	(5) 555-4729
Antonio Moreno Taquería	(5) 555-3932
Around the Horn	(171) 555-7788
Berglunds snabbköp	0921-12 34 65

5.9 Filter by Fax Number Null (Empty)

```
SELECT CompanyName, Fax
FROM Customers
WHERE Fax IS NULL
LIMIT 5;
```

CompanyName	Fax
Antonio Moreno Taquería	
B's Beverages	
Chop-suey Chinese	
Comércio Mineiro	
Familia Arquibaldo	

5.10 Filter by Fax Number Not Null (Not Empty)

```
SELECT CompanyName, Fax
FROM Customers
WHERE Fax IS NOT NULL
LIMIT 5;
```

CompanyName	Fax
Alfreds Futterkiste	030-0076545
Ana Trujillo Emparedados y helados	(5) 555-3745
Around the Horn	(171) 555-6750
Berglunds snabbköp	0921-12 34 67
Blauer See Delikatessen	0621-08924

5.11 Products in Stock Ordered by Price

```
SELECT ProductName, UnitPrice, UnitsInStock
FROM Products
WHERE
UnitsInStock > 0
ORDER BY UnitPrice ASC
LIMIT 5;
```

ProductName	UnitPrice	UnitsInStock
Geitost	2.5	112
Guaraná Fantástica	4.5	20
Konbu	6	24
Filo Mix	7	38
Tourtière	7.45	21

5.12 Products Out of Stock

```
SELECT ProductName, UnitPrice, UnitsInStock
FROM Products
WHERE
UnitsInStock = 0
LIMIT 5;
```

ProductName	UnitPrice	UnitsInStock
Chef Anton's Gumbo Mix	21.35	0
Alice Mutton	39	0
Thüringer Rostbratwurst	123.79	0
Gorgonzola Telino	12.5	0
Perth Pasties	32.8	0

5.13 Products and Their Suppliers

```
SELECT ProductName, SupplierID
FROM Products
LIMIT 5;
```

ProductName	SupplierID
Chai	1
Chang	1
Aniseed Syrup	1
Chef Anton's Cajun Seasoning	2
Chef Anton's Gumbo Mix	2

5.14 Suppliers with Specific IDs

```
SELECT SupplierID, CompanyName, ContactName
FROM Suppliers
WHERE
SupplierID IN (1, 2);
```

SupplierID	CompanyName	ContactName
1	Exotic Liquids	Charlotte Cooper
2	New Orleans Cajun Delights	Shelley Burke

5.15 Products and Their Suppliers - Inner Join

```
SELECT Products.ProductName, Suppliers.CompanyName
FROM Products
INNER JOIN Suppliers
ON Products.SupplierID = Suppliers.SupplierID
LIMIT 5;
```

ProductName	CompanyName
Chai	Exotic Liquids
Chang	Exotic Liquids
Aniseed Syrup	Exotic Liquids
Chef Anton's Cajun Seasoning	New Orleans Cajun Delights
Chef Anton's Gumbo Mix	New Orleans Cajun Delights

5.16 Products and Their Categories - Inner Join

```
SELECT
Products.ProductName, Categories.CategoryName FROM Products
INNER JOIN Categories
ON
Products.CategoryID = Categories.CategoryID
LIMIT 10;
```

ProductName	CategoryName
Chai	Beverages
Chang	Beverages
Aniseed Syrup	Condiments
Chef Anton's Cajun Seasoning	Condiments
Chef Anton's Gumbo Mix	Condiments
Grandma's Boysenberry Spread	Condiments
Uncle Bob's Organic Dried Pears	Produce
Northwoods Cranberry Sauce	Condiments
Mishi Kobe Niku	Meat/Poultry
Ikura	Seafood

5.17 Products with Unit Price Between 10 and 15

```

SELECT ProductID, ProductName, UnitPrice
FROM Products
WHERE
UnitPrice > 10 AND UnitPrice < 15
ORDER BY UnitPrice
LIMIT 5;

```

ProductID	ProductName	UnitPrice
46	Spegesild	12
31	Gorgonzola Telino	12.5
68	Scottish Longbreads	12.5
48	Chocolade	12.75
77	Original Frankfurter grüne Soße	13

5.18 Orders Shipped to Belgium

```

SELECT
OrderID, ShipAddress, ShipCity, ShipCountry
FROM Orders
WHERE
ShipCountry = 'Belgium'
LIMIT 10;

```

OrderID	ShipAddress	ShipCity	ShipCountry
10252	Boulevard Tirou, 255	Charleroi	Belgium
10302	Boulevard Tirou, 255	Charleroi	Belgium
10458	Boulevard Tirou, 255	Charleroi	Belgium
10463	Boulevard Tirou, 255	Charleroi	Belgium
10475	Boulevard Tirou, 255	Charleroi	Belgium
10529	Rue Joseph-Bens 532	Bruxelles	Belgium
10649	Rue Joseph-Bens 532	Bruxelles	Belgium
10760	Rue Joseph-Bens 532	Bruxelles	Belgium
10767	Boulevard Tirou, 255	Charleroi	Belgium
10841	Boulevard Tirou, 255	Charleroi	Belgium

5.19 OR

```

SELECT CompanyName, City from Customers
WHERE
City = 'Madrid' OR City = 'Portland'
ORDER BY City
LIMIT 5;

```

CompanyName	City
Bólido Comidas preparadas	Madrid
FISSA Fabrica Inter. Salchichas S.A.	Madrid
Romero y tomillo	Madrid
Lonesome Pine Restaurant	Portland
The Big Cheese	Portland

5.20 Summary Statistics

```

SELECT * FROM OrderDetails
WHERE
OrderID = 10248;

```

OrderID	ProductID	UnitPrice	Quantity	Discount
10248	11	14	12	0.0
10248	42	9.8	10	0.0
10248	72	34.8	5	0.0

```

SELECT ProductID, SUM(Quantity)
FROM OrderDetails
WHERE
OrderID = 10248

```

ProductID	SUM(Quantity)
11	27

```
SELECT ProductID, SUM(Quantity)
FROM 'Order Details'
GROUP BY ProductID
LIMIT 5;
```

ProductID	SUM(Quantity)
1	828
2	1057
3	328
4	453
5	298

5.21 Aggregate Functions

5.21.1 Maximum Unit Price of Products

```
SELECT max(UnitPrice)
FROM Products;
```

max(UnitPrice)
263.5

5.21.2 Minimum and Maximum Prices of Products in Stock

```
SELECT min(UnitPrice), max(UnitPrice)
FROM Products
WHERE UnitsInStock > 0;
```

min(UnitPrice)	max(UnitPrice)
2.5	263.5

5.21.3 Average Unit Price of Products in Stock

```
SELECT avg(UnitPrice) as 'Average'
FROM Products
WHERE UnitsInStock > 0;
```

Average
27.68430555555556

Outputs the same:

```

SELECT sum(UnitPrice)/count(UnitPrice) as 'Average'
FROM Products
WHERE UnitsInStock > 0;

```

5.21.4 Count of Products in Stock

```

SELECT sum(UnitsInStock)
FROM Products;

```

sum(UnitsInStock)
3119

5.21.5 Distinct Countries of Orders

```

SELECT
DISTINCT(ShipCountry)
FROM Orders;

```

ShipCountry
France
Germany
Brazil
Belgium
Switzerland
Venezuela
Austria
Mexico
USA
Sweden
Finland
Italy
Spain
UK
Ireland
Portugal
Canada

5.21.6 Distinct Countries of Customers

```

SELECT
DISTINCT(Country)
FROM Customers;

```

Country
Germany
Mexico
UK
Sweden
France
Spain
Canada
Argentina
Switzerland
Brazil
Austria
Italy
Portugal
USA
Venezuela
Ireland
Belgium
Norway
Denmark
Finland
Poland

5.22 Regions Table

```
SELECT * FROM Regions;
```

RegionID	RegionDescription
1	Eastern
2	Western
3	Northern
4	Southern

```
SELECT * FROM Regions
WHERE
RegionID < 3
ORDER BY RegionDescription DESC;
```

RegionID	RegionDescription
2	Western
1	Eastern

6 The SUBSTR Function

SUBSTR is a string function that extracts a substring from a given string based on specified starting position and length parameters.

For example SUBSTR('Hello, World!', 1, 5) would return 'Hello', as it starts at position 1 and extracts 5 characters. For example SUBSTR('Hello, World!', 8, 5) would return 'World', as it starts at position 8 and extracts 5 characters. For example SUBSTR('Hello, World!', 8, 6) would return 'World!', as it starts at position 8 and extracts 6 characters.

```
SELECT CompanyName, ContactName, Address, Country
FROM Customers
WHERE
SUBSTR(CompanyName, 1, 2) = 'Co';
```

CompanyName	ContactName	Address	Country
Comércio Mineiro	Pedro Afonso	Av. dos Lusíadas, 23	Brazil
Consolidated Holdings	Elizabeth Brown	Berkeley Gardens 12 Brewery	UK

```
SELECT CompanyName, ContactName, Address, Country
FROM Customers
WHERE
SUBSTR(ContactName, 2, 2) = 'ed';
```

CompanyName	ContactName	Address	Country
Comércio Mineiro	Pedro Afonso	Av. dos Lusíadas, 23	Brazil

```
SELECT CompanyName, ContactName, Address, Country
FROM Customers
WHERE
SUBSTR(Country, 4, 3) = 'zil'
LIMIT 5;
```

CompanyName	ContactName	Address	Country
Comércio Mineiro	Pedro Afonso	Av. dos Lusíadas, 23	Brazil
Familia Arquibaldo	Aria Cruz	Rua Orós, 92	Brazil
Gourmet Lanchonetes	André Fonseca	Av. Brasil, 442	Brazil
Hanari Carnes	Mario Pontes	Rua do Paço, 67	Brazil
Que Delícia	Bernardo Batista	Rua da Panificadora, 12	Brazil

7 The STRFTIME Function

```
SELECT OrderID, CustomerID, OrderDate
```

```
FROM Orders
LIMIT 5;
```

OrderID	CustomerID	OrderDate
10248	VINET	1996-07-04 00:00:00.000
10249	TOMSP	1996-07-05 00:00:00.000
10250	HANAR	1996-07-08 00:00:00.000
10251	VICTE	1996-07-08 00:00:00.000
10252	SUPRD	1996-07-09 00:00:00.000

7.1 Extract Year from OrderDate

```
SELECT
OrderID, CustomerID,
OrderDate, strftime("%Y", OrderDate) as Year
FROM Orders
LIMIT 5;
```

OrderID	CustomerID	OrderDate	Year
10248	VINET	1996-07-04 00:00:00.000	1996
10249	TOMSP	1996-07-05 00:00:00.000	1996
10250	HANAR	1996-07-08 00:00:00.000	1996
10251	VICTE	1996-07-08 00:00:00.000	1996
10252	SUPRD	1996-07-09 00:00:00.000	1996

7.2 Extract Month from OrderDate

```
SELECT
OrderID, CustomerID,
OrderDate, strftime("%m", OrderDate) as Month
FROM Orders
LIMIT 5;
```

OrderID	CustomerID	OrderDate	Month
10248	VINET	1996-07-04 00:00:00.000	07
10249	TOMSP	1996-07-05 00:00:00.000	07
10250	HANAR	1996-07-08 00:00:00.000	07
10251	VICTE	1996-07-08 00:00:00.000	07
10252	SUPRD	1996-07-09 00:00:00.000	07

7.3 Extract Day from OrderDate

```
SELECT
```



```

OrderID, CustomerID,
OrderDate, strftime("%d", OrderDate) as Day
FROM Orders
LIMIT 5;

```

OrderID	CustomerID	OrderDate	Day
10248	VINET	1996-07-04 00:00:00.000	04
10249	TOMSP	1996-07-05 00:00:00.000	05
10250	HANAR	1996-07-08 00:00:00.000	08
10251	VICTE	1996-07-08 00:00:00.000	08
10252	SUPRD	1996-07-09 00:00:00.000	09

8 The LENGTH Function

```

SELECT
    ProductName,
    UnitPrice
FROM Products
WHERE LENGTH(ProductName) > 30
LIMIT 5;

```

ProductName	UnitPrice
Uncle Bob's Organic Dried Pears	30
Jack's New England Clam Chowder	9.65
Louisiana Fiery Hot Pepper Sauce	21.05
Original Frankfurter grüne Soße	13

9 LOWER and UPPER Functions

```

SELECT
    LOWER(ProductName) AS 'L',
    UPPER(ProductName) AS 'U',
    UnitPrice
FROM Products
WHERE
LENGTH(ProductName) > 30
LIMIT 5;

```

L	U	UnitPrice
uncle bob's organic dried pears	UNCLE BOB'S ORGANIC DRIED PEARS	30
jack's new england clam chowder	JACK'S NEW ENGLAND CLAM CHOWDER	9.65
louisiana fiery hot pepper sauce	LOUISIANA FIERY HOT PEPPER SAUCE	21.05
original frankfurter grüne soße	ORIGINAL FRANKFURTER GRÜNE SOßE	13

10 LTRIM, RTRIM, and TRIM Functions

```
SELECT
LTRIM('    Text ') as 'L',
RTRIM('    Text ') as 'R',
TRIM ('    Text ') as 'T'
FROM Products
LIMIT 1;
```

L	R	T
Text	Text	Text

Note: Since it's not clear in the output,

1. **LTRIM** removes the leading spaces from the string (Removes spaces from the left).
2. **RTRIM** removes the trailing spaces from the string (Removes spaces from the right).
3. **TRIM** removes both leading and trailing spaces from the string (Removes spaces from both sides).

11 The INSTR Function

```
SELECT
ProductName
FROM Products
WHERE INSTR(ProductName, 'ea') > 0
LIMIT 5;
```

ProductName
Chef Anton's Cajun Seasoning
Grandma's Boysenberry Spread
Uncle Bob's Organic Dried Pears
Teatime Chocolate Biscuits
Boston Crab Meat

Note: The **INSTR** function returns the position of the first occurrence of a substring within a string. In the example above, it checks if the substring 'ea' exists in the ProductName and returns its position. If the position is greater than 0, it means the substring is found in the ProductName.

12 UPDATE

Description: Updates the unit price values of the products with product IDs 1, 2, 3, 4, and 5 by multiplying their current unit price values by 1.25.

12.1 Update Unit Prices of Specific Products

```
UPDATE Products
SET UnitPrice = UnitPrice * 1.25
WHERE ProductID IN (1, 2, 3, 4, 5);
```

12.2 Increase Units in Stock of Specific Products

Description: Increases the units in stock values of the products with product IDs 1 and 2 by adding 10 to their current units in stock values.

```
UPDATE Products
SET
UnitsInStock = UnitsInStock + 10
WHERE ProductID IN (1, 2);
```

12.3 Update Customer Information

Description: Updates the city and phone number of the customer with a customer ID of 1 to 'İstanbul' and '(90) 212 123 45 67', respectively.

```
UPDATE Customers
SET
City = 'İstanbul',
Phone = '(90) 212 123 45 67'
WHERE
CustomerID = 1;
```

13 DELETE

13.1 Delete Specific Products

Description: Deletes the products with product IDs 1, 2, 3, 4, and 5 from the Products table.

```
DELETE FROM Products
WHERE ProductID IN (1, 2, 3, 4, 5);
```

14 INSERT INTO

14.1 Insert a New Product

Description: Insert a new product with the name 'New Product', a supplier ID of 1, a category ID of 1, a unit price of 9.99, and 100 units in stock into the Products table.

```
INSERT INTO Products
(ProductName, SupplierID, CategoryID, UnitPrice, UnitsInStock)
VALUES
('New Product', 1, 1, 9.99, 100);
```

14.2 Insert a New Customer

Description: Insert a new customer with the name 'New Customer', a contact name of 'John Doe', a city of 'New York', and a country of 'USA' into the Customers table.

```
INSERT INTO Customers
(CustomerName, ContactName, City, Country)
VALUES
('New Customer', 'John Doe', 'New York', 'USA');
```

15 DROP TABLE

15.1 Delete Products Table

Description: Deletes the Products table from the database if it exists.

```
DROP TABLE IF EXISTS Products;
```

16 CASE - WHEN

```
SELECT
ProductName, UnitPrice,
CASE
    WHEN UnitPrice <= 20 THEN 'Low Price'
    WHEN UnitPrice > 20 AND UnitPrice <= 40 THEN 'Moderate'
    WHEN UnitPrice > 40 THEN 'High Price'
END AS 'Price Description'
FROM Products
LIMIT 10;
```

ProductName	UnitPrice	Price Description
Chai	18	Low Price
Chang	19	Low Price
Aniseed Syrup	10	Low Price
Chef Anton's Cajun Seasoning	22	Moderate
Chef Anton's Gumbo Mix	21.35	Moderate
Grandma's Boysenberry Spread	25	Moderate
Uncle Bob's Organic Dried Pears	30	Moderate
Northwoods Cranberry Sauce	40	Moderate
Mishi Kobe Niku	97	High Price
Ikura	31	Moderate

17 Exercises

Suppose that Prices table contains the following fields: ProductID, UnitPrice, and UnitsInStock. Here is the table data:

ProductID	UnitPrice	UnitsInStock
1	10	100
2	20	200
3	30	300
4	40	400
5	50	500

Write SQL queries to perform the following operations:

- Calculate the total value of the inventory (UnitPrice multiplied by UnitsInStock for each product, and then summed across all products).
- Find the average unit price of the products in stock.
- List the products that have a unit price greater than 25 and are in stock.
- Update the unit price of all products by increasing it by 10
- Delete the products that have a unit price less than 15.

Answers:

```
-- Total value of the inventory
SELECT SUM(UnitPrice * UnitsInStock) AS TotalInventoryValue
FROM Prices;
```

```
-- Average unit price of the products in stock
SELECT AVG(UnitPrice) AS AverageUnitPrice
FROM Prices
WHERE UnitsInStock > 0;
```

```
-- List products with unit price greater than 25 and in stock
SELECT ProductID, UnitPrice, UnitsInStock
FROM Prices
WHERE UnitPrice > 25 AND UnitsInStock > 0;

-- Update unit price by increasing it by 10%
UPDATE Prices
SET UnitPrice = UnitPrice * 1.10;

-- Delete products with unit price less than 15
DELETE FROM Prices
WHERE UnitPrice < 15;
```

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